Should the Gilded Ace replace its ten lowest-earning slot machines? We built an interactive dashboard using six months of casino data to investigate that decision. We compared each machine’s actual winnings with its expected winnings per available hour. This accounts for differences in operating time and helps managers distinguish weak demand from an unusual run of payouts. Several low-performing machines were located in walkways, making placement worth testing before buying replacements.

Demand varied substantially by day. Saturday wagers averaged 1.80 million dollars, compared with 1.10 million on Tuesdays. We trained a regression model on January through April and tested it on May and June. Using the weekday reduced the average daily forecast error from 218,258 dollars to 68,699 dollars. Adding holiday status improved it by only another 346 dollars per day. Tuesday is therefore a reasonable starting point for scheduling routine maintenance.

Machines accumulated 5,021 hours of downtime. Using each machine’s wagering history during comparable operating hours, we estimated 1.09 million dollars in missed machine wagers. The 9–10 p.m. window accounted for the largest total. The casino-wide impact depends on whether guests continued playing elsewhere on the floor.

The dashboard also shows what guests experienced. Among 96,559 active tracked players, the median player finished 96 dollars behind. Among those who finished behind, the tenth with the largest losses accounted for 52.84% of losses. They wagered 13.3 times as much per person while losing a smaller share of each dollar wagered. Loyalty rewards cost the casino 190,669 dollars over the same period, about 0.81% of expected house winnings.

We recommend a 30-day pilot: spend 1,800 dollars to move one low-performing walkway machine and reserve the remaining 98,200 dollars. Compare its expected winnings per available hour before and after the move against a similar machine that stayed in place. Track neighboring machines as well: a gain at the new location is more valuable if total floor play also increases.